The Big Pool Bangs Theory

The Big Pool Bangs Theory: Cosmology via Jammed Phase Transitions and the Hard-Sphere Kinetic Bounce

Abstract. We explore a mechanical alternative to inflationary ΛCDM in which the universe begins as a jammed ensemble of $\sim 10^{60}$ Planck-mass relics, treated as effective hard spheres (an ansatz motivated by the Compton–Schwarzschild coincidence).[1] A loop-quantum-cosmology-type modified Friedmann equation, with critical density conjecturally identified with the hard-sphere jamming density, produces a non-singular bounce; the subsequent chaotic rebound ejects most of the ensemble. Toy N-body simulations with a turbulent kick distribution eject $88\% \pm 4\%$ of the mass, consistent with the observed dark-matter fraction for $\sigma_{\rm rel} \approx 1$ — a regime a cascade nucleation model only marginally reaches — and show clean velocity sorting of the ejecta. The framework does *not* reproduce cosmic acceleration: coasting expansion is disfavored by Pantheon+ ($\Delta\chi^2 \approx 84$) and DES-SN5YR ($\Delta\chi^2 \approx 131$), a relativistic LTB lightcone through the ejecta decelerates ($q_0^{\rm eff} = +0.16$), a toy effective $w(z)$ drifts in the wrong direction relative to the DES+DESI fit, and a kinetic-Sunyaev–Zel'dovich gate shows that no ejection profile can supply more than $\sim 1\%$ of the observed dark energy. We therefore adopt the pool model *plus a constant tension of space* ($w = -1$) as the expansion history: the rebound sets the origin and velocity structure of the matter, the tension sets the acceleration, and the value of the tension is not derived. A predicted Cold Spot hot ring at 5ř–7ř is absent in Planck 2018 data. Because a neighbouring universe imprinting the last-scattering surface lies beyond our particle horizon, only an initial-condition (Sachs–Wolfe) imprint of its potential is causally admissible; for the Cold Spot this fixes a source mass $M_2 \approx 10^{48}$ kg (± 0.5 dex) 1–2.5 Gly beyond last scattering, whose tidal pull on the local volume is $\lesssim 1$ km s^{-1}. We organize the claims into seven hypotheses[2] — the bounce-and-ejection core (H1), the Cold Spot as a frozen vortex (H2), a neighbouring Big Pool Bang in the Cold Spot direction (H3), a population of such neighbours (H4), our own position inside Universe 1 (H5), the location of the second Big Pool Bang (H6), and the population model (H7) — each with explicit falsifiers collected in one table, catalogue ten predictions (P1–P10) with the instruments required, and state the unsolved problems: causal structure of the bounce, relativistic formulation of ejection, the relativistic (hot) temperature of the ejecta, the horizon and flatness problems, and the expansion history.

1. Introduction: Three Questions

This paper is organized around three questions.

[1] Code, figures, build instructions and download links for all datasets used: https://github.com/AIMFIRST-VN/big001

[2] The author's working nicknames for the hypotheses, retained here for continuity with earlier drafts: H1 "The Big Pool Bang"; H2 "The Cold Eye of Sauron"; H3 "Big Pool Bang Multiverse 2"; H4 "The Dark Comic Background"; H7 "The Pool Model". Headings use plain titles.

1. **Where are we?** If the universe began as a jammed ensemble of Planck-mass relics that rebounded and ejected most of its mass, the ejection was velocity-sorted (Sec. 3, Fig. 2). In the cosmological setting the jammed patch is the whole universe and ejection happens everywhere, so the question is whether any residual large-scale radial ordering survives in our Hubble volume. Our current answer (H5, Sec. 10): the displaced-centre test is null to within a few Mpc, the bulk flow at $\sim$400 km s^{-1} is the only kinematic handle, and the hemispherical asymmetry is better assigned to an external neighbour than to our own offset.

2. **Where is the second Big Bang?** A neighbouring Big Pool Bang beyond our particle horizon can imprint the last-scattering surface only through the potential it contributed to the initial conditions — a Sachs–Wolfe spot with a low-multipole tail. The Cold Spot at $(l, b) = (203\text{ř}, -56\text{ř})$ is the candidate (H2, H3, H6, Secs. 6–8): direction known to $\pm 3\text{ř}$, comoving distance 1–2.5 Gly beyond the last-scattering surface, mass $M_2 \approx 10^{48}$ kg to within ± 0.5 dex, and a curl-mode test (P8) pre-registered. The eight extreme spots of the Planck map are the candidate list for a population of such neighbours (H4, H7, Sec. 9).

3. **How does the Big Pool Bang model work, and what does it explain?** A first-order vacuum crystallization jams into a hard-sphere state; a loop-quantum-cosmology-type bounce rebounds it; chaotic ejection sends $88\% \pm 4\%$ of the mass outward as dark matter (H1, Secs. 2–4). The rebound kinetics do *not* reproduce cosmic acceleration, so the adopted expansion history is the pool model plus a constant tension of space whose value is not derived (Sec. 5.2); the predicted Cold Spot thermal ring is absent, and the ejecta are relativistic, so "cold" dark matter is unproven (Sec. 12).

The paper answers the third question first, then treats the Cold Spot (Sec. 6), the neighbour mechanism (Sec. 7), its location (Sec. 8), the population (Sec. 9), and our own position (Sec. 10). Section 11 collects the ten predictions and the single falsification table; Sec. 12 lists open problems.

1.1 Vacuum Crystallization and the Hard-Sphere State

Standard cosmology relies on scalar-field inflation to explain why the observable universe is flat, isotropic, and thermally uniform, at the cost of fine-tuning and of singularity theorems (Borde–Guth–Vilenkin). The **Big Pool Bangs Theory** is a mechanical alternative grounded in first-order phase-transition dynamics. Space begins in a false vacuum with $\rho_{\rm vac} \sim \rho_P$. Quantum fluctuations trigger a **vacuum crystallization** transition which nucleates at many points: bubbles of converted phase grow, collide, and merge, and the "bang" is their percolation into a single jammed state.

Timeline. Bubble walls propagate at $\leq c$, so percolating a $\sim 10^{20}\,\ell_P$ patch takes at least $10^{20}\,t_P = 5.4 \times 10^{-24}$ s divided by the number of nucleation sites per crossing length; any nucleation density sparse enough to give nontrivial merger statistics implies an assembly time many orders above t_P, whose consistency with trapped-surface avoidance (Sec. 3) is open.

Nucleation as the origin of the kick spectrum. The randomness of the merger history is inherited by the jammed core as internal turbulence, a candidate source for the stochastic kick spectrum that Sec. 3 otherwise requires as an input — provided nucleation-era velocity structure survives the collisional jammed epoch rather than thermalizing. A Poisson nucleation-and-merger simulation (vector-summed wall impulses) recovers a Maxwellian kick distribution (speed coefficient of variation 0.41–0.42), largely a central-limit consequence; the nontrivial statistic is the coherent-to-dispersive ratio $\sigma_{\rm rel}$. Poisson nucleation is turbulence-dominated ($\sigma/\langle v_r \rangle \approx 5$–8) and cannot supply the coherent radial component the ejection fraction requires. A *cascade* model — a single seed bubble whose wall catalyzes further nucleation, halted by a jamming feedback at $\eta \approx 0.64$ — yields $\sigma_{\rm rel} = 0.65$–0.78, robust across a 4× range in trigger distance, against the ≈ 0.75–1.0 window the dark-matter fraction requires: the cascade reaches that window only marginally. Three caveats gate the chain: the $88\% \pm 4\%$ figure comes from Maxwellian-ansatz N-body runs and the cascade's heavier-tailed kick field has not been fed through the N-body pipeline; the model retains structural choices (trigger distance, nucleation lag, wall-area weighting, the arrest $\eta = 0.64$); and the wall-momentum efficiency is assumed universal, whereas in first-order-transition physics it is model-dependent and typically small (Espinosa–Konstandin–No–Servant). Bubble collisions in any first-order transition also source a stochastic gravitational-wave background peaked at the transition scale; this is a generic prediction of H1 (P3), not of any neighbour.

Because quantum geometry prohibits localized infinite densities, the energy crystallizes into $N \approx 10^{60}$ irreducible **Planck-mass relics** ($m_P \approx 21.8\,\mu$g), total mass $M_1 = N m_P \approx 2.2 \times 10^{52}$ kg (N is order-of-magnitude; for comparison, the matter mass of the *observable* universe is $\sim 1.5 \times 10^{53}$ kg). The relics saturate local volume up to the random close-packing limit $\eta_{\rm rcp} \approx 0.64$. With $\rho_P = m_P/\ell_P^3 = 5.2 \times 10^{96}$ kg m^{-3} the nominal jamming radius would be

$$R_{\rm jam} = \left(\frac{3M_1}{4\pi\rho_P\eta_{\rm rcp}}\right)^{1/3} \approx 1.2 \times 10^{-15} \text{ m},$$

but this must not be read as the size of an isolated object: the Schwarzschild radius of 2.2×10^{52} kg is 3.3×10^{25} m, forty orders larger, so a finite ball of this mass and density would be a black hole, not a bouncing patch. The bounce of Sec. 3 is meaningful only if the jammed patch is the whole (or a super-horizon) universe described by a homogeneous FLRW-type metric, for which no exterior Schwarzschild geometry exists. We therefore use ρ_c, not $R_{\rm jam}$, as the physical parameter.

2. Planck-Mass Relics as Effective Hard Spheres

For a particle of mass m the reduced Compton wavelength $\lambda_C = \hbar/mc$ and the Schwarzschild radius $R_s = 2Gm/c^2$ coincide at $m \approx m_P$, where both equal $\sim \ell_P = 1.616 \times 10^{-35}$ m. This is the regime in which neither quantum field theory nor classical general relativity is separately valid. **We adopt, as a modeling ansatz rather than a derivation, that Planck relics possess a minimum spatial extent $\sim \ell_P$ that no interaction can compress**, and treat them as effective incompressible hard spheres. The ansatz is judged by the phenomenology it generates (Secs. 3–10); its microphysical justification is an open problem (Sec. 12).

3. The Kinetic Bounce and Chaotic Ejection (Hypothesis 1)

Why hard-sphere pressure cannot bounce the collapse. Inside a homogeneous fluid the dynamics are governed by the Raychaudhuri equation

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}\left(\rho + \frac{3P}{c^2}\right).$$

Near random close packing the hard-sphere kinetic pressure diverges (free-volume equation of state, $P \propto nk_BT\,[1 - (\eta/\eta_{\rm rcp})^{1/3}]^{-1}$), but in general relativity positive pressure *gravitates*: no classical equation of state with $P > 0$ can bounce an FLRW collapse. The rebound must be sourced by quantum-geometric corrections that dominate at the jamming density $\rho_c = \eta_{\rm rcp}\,\rho_P$. The minimal object with the required property is the loop-quantum-cosmology (LQC) modified Friedmann equation,

$$H^2 = \frac{8\pi G}{3}\,\rho\left(1 - \frac{\rho}{\rho_c}\right), \qquad \rho_c = \eta_{\rm rcp}\,\rho_P,$$

which encodes a maximum attainable density covariantly: $H \to 0$ as $\rho \to \rho_c$ and the effective Raychaudhuri equation flips sign. We propose the jamming limit of the relic gas as the physical origin of the ρ_c that LQC introduces axiomatically, noting that LQC's own Immirzi-fixed value is $\rho_c \approx 0.41\,\rho_P$ versus $0.64\,\rho_P$ here — order-of-magnitude agreement, a coincidence rather than a derivation. A numerical integration (Fig. 1) confirms that the classical hard-sphere fluid overshoots the jamming density by $> 10^4$ while the modified equation bounces at $\rho = \rho_c$; the bounce is built into the equation, not independently confirmed by it. Whether it completes before a trapped surface forms around the still-homogeneous patch is an assumption (Sec. 12).

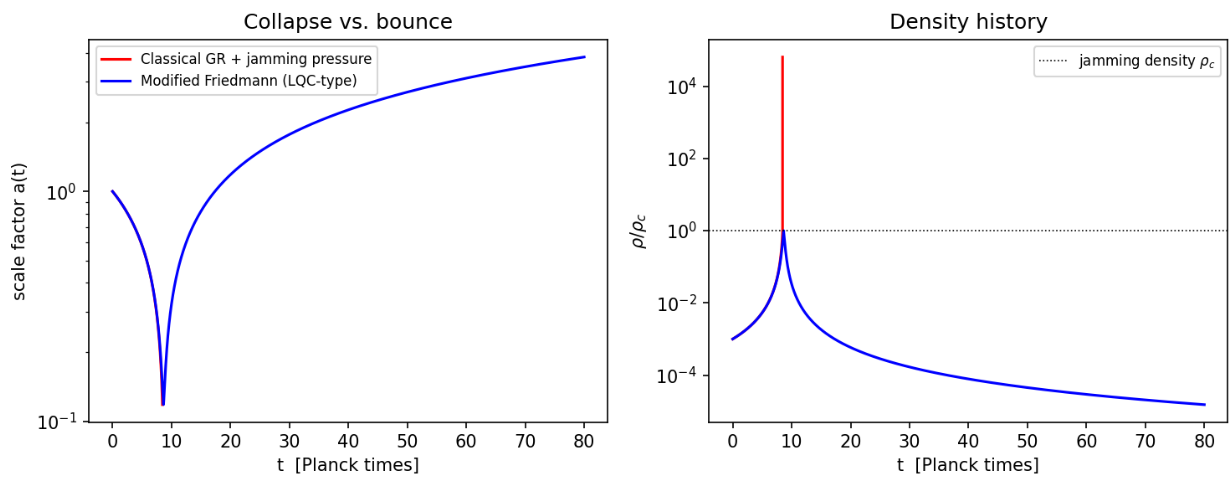

Figure 1: Fig. 1 — Toy FLRW integration. Left: scale factor. Right: density history. Classical GR with jamming pressure (red) overshoots the jamming density by $> 10^4$ (collapse); the modified Friedmann equation (blue) bounces at ρ_c.

Regime disclaimer. Everything that follows in this section is a **Newtonian analogy**: the real dynamics occur where escape velocities formally exceed c by $\sim$20 orders of magnitude and only a general-relativistic treatment is meaningful. We use the Newtonian model because violent-relaxation statistics (which fraction unbinds, how ejecta sort by speed) are plausibly governed by dimensionless energy ratios that survive the translation; whether they do is the central open problem of Sec. 12.

The slingshot phase. Picture the jammed core as 10^{60} billiard balls packed to the jamming limit; when the bounce reverses the collapse, the rebound is the break. At jamming density the relic gas is collisional, so the slingshot phase begins only after expansion dilutes the ensemble below the contact threshold and relics decouple onto ballistic trajectories. From then on the system is a dense self-gravitating N-body ensemble in which three-body encounters dominate relaxation — the "core collapse and halo ejection" of globular-cluster dynamics, which sheds most of a system's mass to reach a bound remnant. Applied to the birth of the universe it sorts the mass into two components:

- **The ejected halo ($\approx 88\%$):** relics kicked above escape velocity, free-streaming and rarely colliding again — the candidate dark-matter component, requiring no new particle (but see the temperature caveat below).

- **The trapped furnace ($\approx 12\%$):** relics that fall back and grind against each other in a turbulent core, proposed as the origin of the photon–baryon plasma. **Caveat (essential):** this requires an unspecified conversion channel from Planck relics to Standard-Model plasma — including baryogenesis and the photon-to-baryon ratio — while ejected relics remain stable for 13.8 Gyr. Absent such a channel (e.g. a threshold collision energy the ejecta never reach), the trapped fraction cannot be identified with the baryon budget.

Ejecta temperature. The ejecta leave at a substantial fraction of the (formally superluminal) escape speed, i.e. relativistically. Without a redshifting or cooling argument that brings their thermal velocities below the free-streaming limit by the epoch of structure formation, Planck-relic ejecta behave as *hot*, not cold, dark matter. Cosmological redshifting of free-streaming momenta ($p \propto 1/a$) is the natural candidate, but no calculation of the residual dispersion at matter–radiation equality exists here; this is a first-order open problem (Sec. 12).

Ejection fraction (toy N-body). A softened $N = 1000$ equal-mass simulation of a cold uniform sphere given a radial kick $v = f\, v_{\rm esc}$ has a steep ejection threshold: $f = 0.6 \to 3\%$, $0.8 \to 41\%$, $0.9 \to 98\%$, $f \geq 1 \to 100\%$. A monolithic kick reproduces the dark-matter fraction only in the narrow window $f \approx 0.85$–0.9, and the zero-energy bounce ($f = 1$) over-ejects. A Maxwellian turbulent component of dispersion $\sigma_{\rm rel}\bar{f}v_{\rm esc}$ superimposed on a mean rebound $\bar{f}v_{\rm esc}$ broadens the transition: at $\bar{f} = 0.7$ the ejected fraction rises from 22% ($\sigma_{\rm rel} = 0.5$) through $\approx 88\%$ (1.0) to 98% (1.5). Over a 15-seed ensemble at $(\bar{f}, \sigma_{\rm rel}) = (0.7, 1.0)$ the ejected fraction is $88\% \pm 4\%$, consistent with the observed dark-matter mass fraction, at a $\sigma_{\rm rel}$ the cascade model of Sec. 1 only marginally reaches; $(\bar{f}, \sigma_{\rm rel})$ remain inputs to be derived from the bounce turbulence spectrum.

Ejecta kinematic structure (15 seeds, 13{,}140 ejecta). Ejection is extended (median ejection time $\sim$1 dynamical time; 10% of ejecta leave after 11), with clean monotonic velocity sorting: median speeds fall from $0.71\, v_{\rm esc}$ for the earliest ejecta to $0.28\, v_{\rm esc}$ for the latest (Fig. 2), so radial position maps onto ejection epoch. The tangential structure shows no correlated anisotropy: the per-sky ejection dipole (0.064 ± 0.024) matches shot noise (0.053 ± 0.024), and early-vs-late shell axes align only weakly ($\langle \cos\theta \rangle = 0.37 \pm 0.49$). The isotropic cascade therefore supplies no preferred axis; if one exists it must be seeded externally (Sec. 7).

From the N-body cloud to the homogeneous universe. The N-body ejecta cloud is a sub-horizon toy of the ejection *mechanism*, not a map of the universe. In the cosmological setting the jammed patch is the whole (super-horizon) universe (Sec. 1), there is no exterior, and ejection happens everywhere: every comoving region is simultaneously "furnace" and "halo". The radial shells of Fig. 2 therefore translate into *local velocity-space structure* within a homogeneous background, not into a single centred cloud with an edge. What survives the translation is the statistical content (ejected fraction, speed distribution, isotropy); what does not is the geometric centre. Whether any residual large-scale radial ordering nevertheless survives in our Hubble volume is the question posed to H5 (Sec. 10).

4. Consistency Checks

- **Horizon problem — open.** The jammed patch is $\sim 10^{20}$ Planck lengths across and cannot equilibrate within a few Planck times; homogeneity of the initial patch is an assumed initial condition.

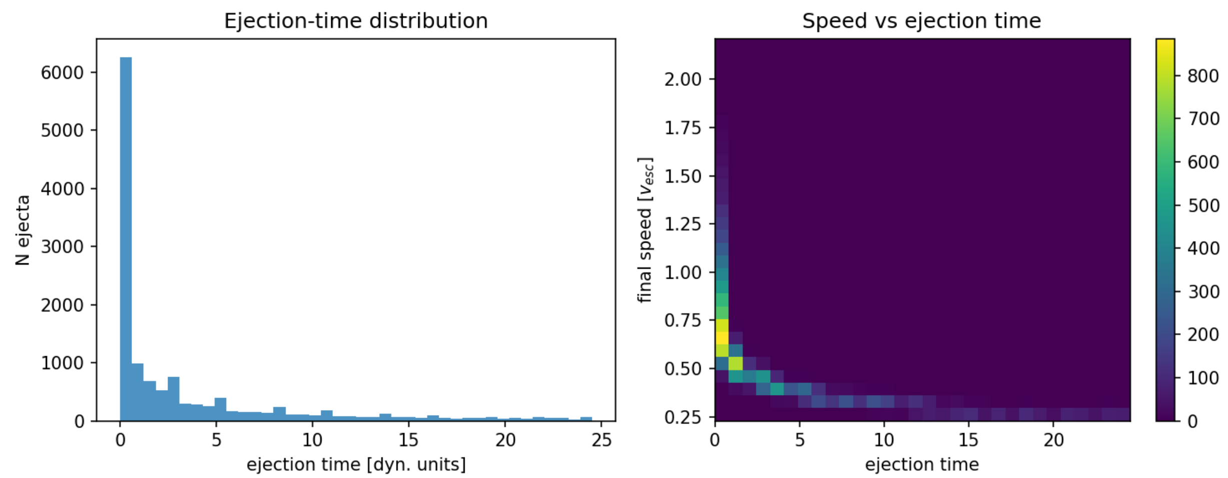

Figure 2: Fig. 2 — Ejecta kinematics over 15 seeds: ejection-time distribution (left) and speed vs ejection time (right), showing the velocity-sorted shell structure.

- **Flatness — consistent, not explained.** In the turbulent regime the simulations favor ($\bar{f} = 0.7$, $\sigma_{\text{rel}} = 1.0$) the mean kinetic energy is $\langle v^2 \rangle = \bar{f}^2(1 + \sigma_{\text{rel}}^2)v_{\text{esc}}^2 \approx 0.98\, v_{\text{esc}}^2$: total energy $K + U \approx 0$ *while* ejecting $\sim$88%, the positive energy of the ejecta balanced by the binding energy of the furnace. Why the rebound selects $E \approx 0$ is not derived.

Remark on the spectral tilt. One may parametrize the scalar index by a turbulent dissipation ratio ν_t (eddy turnover time to expansion time) as $n_s - 1 = -\nu_t/3$; Planck's $n_s = 0.965 \pm 0.004$ then gives $\nu_t \approx 0.105$. Both the relation and the value are fitted inputs, and showing that a Kolmogorov cascade ($E(k) \propto k^{-5/3}$) can produce a nearly scale-invariant curvature spectrum is an open problem (Sec. 12).

5. Expansion History: Why the Pool Model Needs a Constant Tension

The rebound kinetics alone imply a coasting expansion ($a \propto t$), and this is excluded. Against 1371 Hubble-flow SNe Ia from Pantheon+ (diagonal errors, offset marginalized), coasting yields $\chi^2 = 674$ vs 590 for flat ΛCDM ($\Delta\chi^2 \approx 84$), though far better than Einstein–de Sitter ($\chi^2 = 1170$; Fig. 3); on DES-SN5YR (Dovekie recalibration, 1820 SNe, full covariance) the figures are 1632, 1763 ($\Delta\chi^2 \approx 131$), and 2566. Whether the *distribution* of the ejecta could mimic acceleration for an interior observer is tested in Appendix C: a relativistic Lemaître–Tolman–Bondi lightcone through the measured ejecta shells decelerates ($q_0^{\text{eff}} = +0.16$); no smooth, bimodal, or lumpy ejection profile accelerates; a blast-wave cavity is the giant-void geometry; and the kinetic Sunyaev–Zel'dovich limit on coherent cluster flows ($v \lesssim 300$ km s^{-1} at Gpc distances) caps any cavity's contribution at $|\Delta q_0| \lesssim 0.01$ against the $\Delta q_0 \approx -1.05$ required. **No ejection profile of any morphology supplies more than $\sim$1% of the observed dark energy.**

5.1 The $w(z)$ drift

Integrating the measured ejecta speed distribution as cold spherical shells decelerating in their enclosed mass, and computing the equation of state an interior observer would fit, gives w_{eff} drifting from -0.29 at $z = 2$ to -0.31 today (Fig. 4): a few-percent drift near the coasting value $-1/3$.

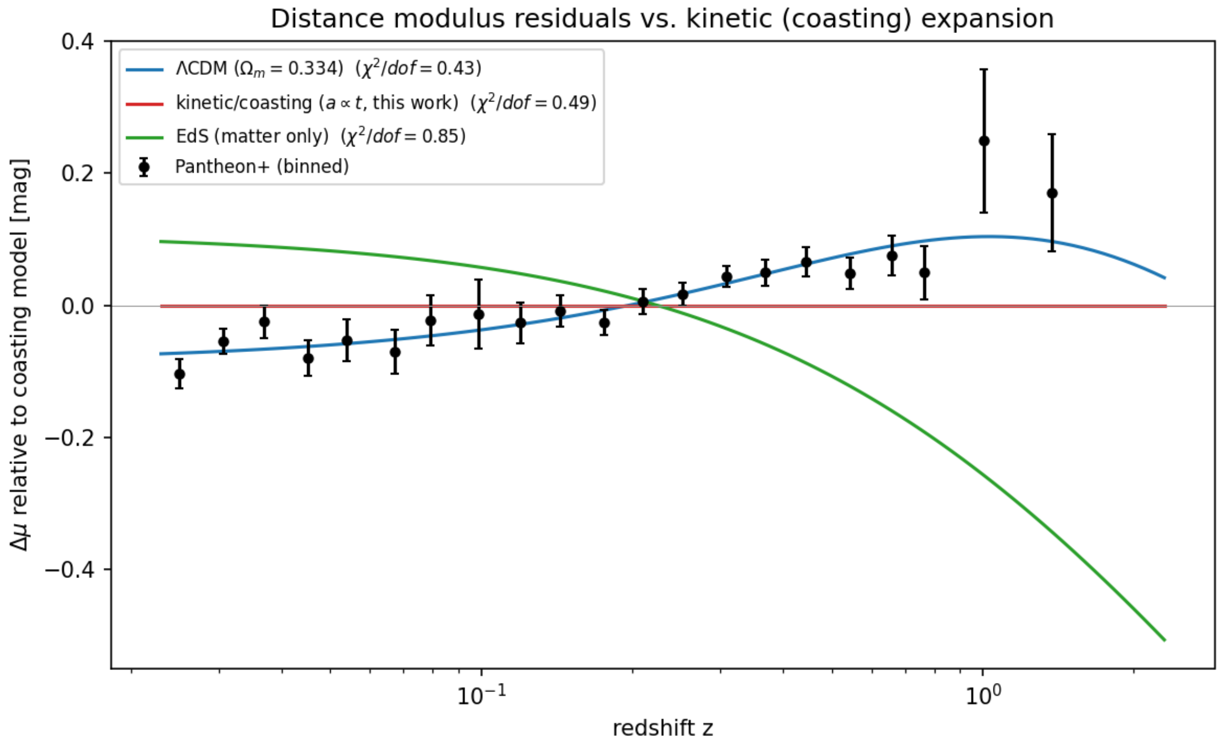

Figure 3: Fig. 3 — Pantheon+ binned distance-modulus residuals relative to the coasting model.

Because this apparent equation of state is set by the evolving ratio of velocity-dispersion energy to clustered rest-mass energy, it cannot be constant. The DES-SN5YR + DESI DR2 BAO $w_0 w_a$CDM fit prefers evolution at 3.2σ ($w_0 = -0.803 \pm 0.054$, $w_a = -0.72 \pm 0.21$), with w becoming *less* negative with time; the toy shells drift the *opposite* way. The remap is crude (91% of the ejecta re-bind in a static enclosed-mass model, so the flow is built from the fastest 9%), but as it stands the kinetic sector shares with the data only the qualitative statement that w evolves, and predicts the wrong sign of the drift.

5.2 The adopted expansion history: the pool model plus a constant tension

The failures above are failures of one idea only: that the *kinetics* of the rebound could substitute for dark energy. They do not touch the bounce, the ejection, or the dark-matter fraction. We therefore separate the two sectors. The pool model is responsible for the origin and velocity structure of the matter; the acceleration is attributed to a constant tension of space, $\rho_\Lambda c^2 = -p_\Lambda$, which we adopt without claiming to derive it.

Why a tension. For a comoving separation $d = a\chi$, $\ddot{d} = (\ddot{a}/a)\, d$ with $\ddot{a}/a = -\frac{4\pi G}{3}(\rho + 3p/c^2)$: acceleration is proportional to distance, with a coefficient set locally by density and pressure, and is repulsive only for $\rho + 3p/c^2 < 0$. If space stores an energy density $u(a)$ when stretched, thermodynamics ($p = -\partial(uV)/\partial V$) fixes its pressure: for $u \propto a^{-n}$, $w = n/3 - 1$. Matter ($n = 3$) attracts; strings or curvature ($n = 2$, $w = -1/3$) coast; a domain-wall *network* ($n = 1$, $w = -2/3$) repels; a constant tension ($n = 0$, $w = -1$) repels; an energy growing as a^2 ($w = -5/3$) is phantom. The wall-network value applies to many walls per Hubble volume; a single wall at $\sim$46 Gly contributes no such term. Observationally, Planck+BAO+SN constrain a constant equation of state

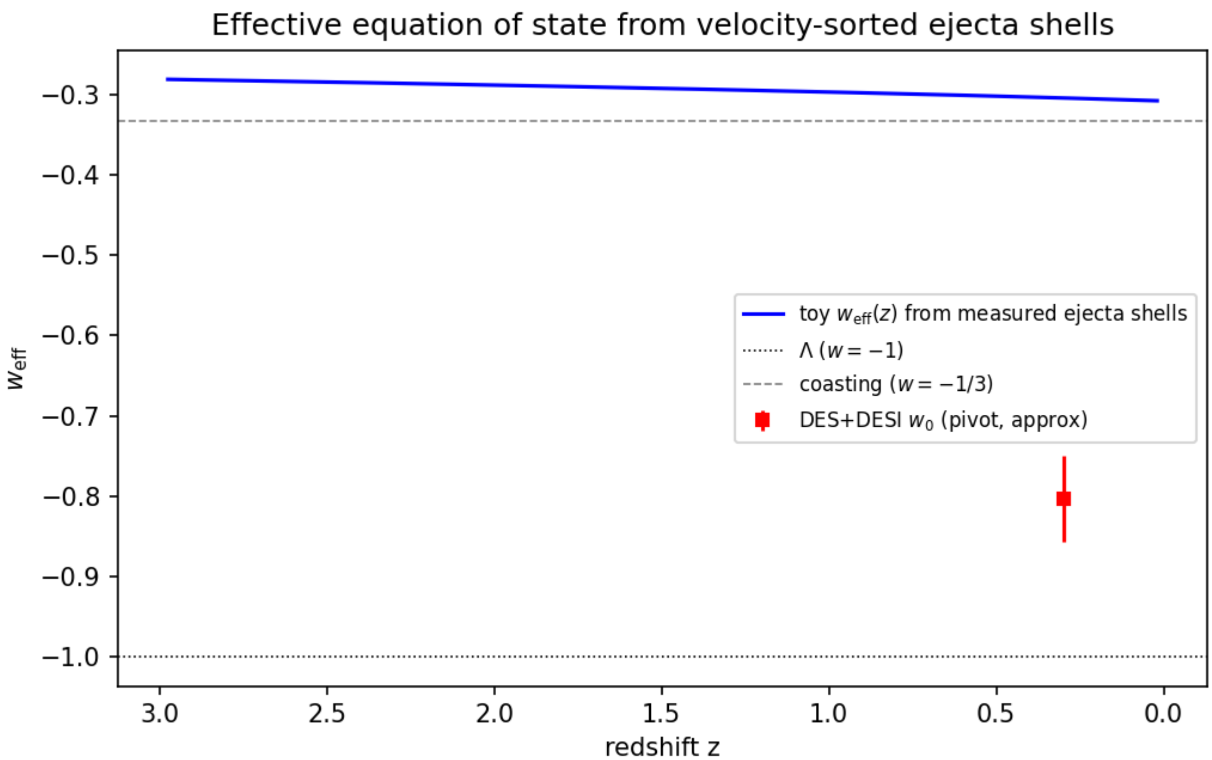

Figure 4: Fig. 4 — Toy effective equation of state from velocity-sorted ejecta shells: a few-percent drift near $w = -1/3$, in the opposite direction to the DES+DESI w_0w_a best fit, and of insufficient magnitude.

to $w = -1.03 \pm 0.03$, while the DES+DESI $w_0 w_a$ fit prefers evolution at 3.2σ (Sec. 5.1). We adopt constant $w = -1$ as the minimal assumption consistent with the first, noting that the evolving-w hint, if confirmed, is not something the kinetic sector can explain: its toy drift has the wrong sign. Neighbouring Big Pool Bangs cannot contribute: a single neighbour supplies a tidal quadrupole and a negligible bulk velocity (Sec. 8), a symmetric population supplies nothing by the shell theorem (Appendix B), and neither is a monopole. Today the tension term gives $\ddot{a}/a = +3.4 \times 10^{-36}$ s^{-2} against -0.7×10^{-36} s^{-2} from matter.

Vacuum energies of neighbours do not sum. Vacuum energy acts only where it resides: the expansion rate of our region is fixed by the tension inside it, so the tensions of any number of neighbouring bangs contribute nothing here, for the same locality reason that their masses contribute no monopole (Appendix B). If all bangs crystallized into the same vacuum, the tension is one universal constant whose value the model does not predict; if into different vacua, the differences reside in the walls between them, and a higher-tension neighbour drives its wall outward as a bubble collision.

The background is assumed, not derived. With the tension included, the ejecta — once velocity-sorted into a Hubble flow and virialized into halos — are treated as pressureless dust on a flat ΛCDM background; the $w_{\rm eff} \approx -0.3$ of the shell toy (Sec. 5.1) describes the shell *kinematics*, not the background equation of state. The supernova, BAO and kSZ comparisons are then satisfied because the ΛCDM background is assumed, and the ejecta shells of Sec. 3 describe how the matter component acquired its Hubble flow rather than an alternative expansion law. What the framework does not supply is the value of the tension: the one framework-native candidate, a residual unconverted vacuum fraction, reproduces the 10^{-123} tuning of standard cosmology (Appendix C), and the origin of ρ_Λ is counted among the open problems (Sec. 12), exactly as in ΛCDM.

6. H2: The Cold Spot and the Axis of Evil

The Eridanus Cold Spot is the sky's outstanding non-Gaussian feature. On the Planck SMICA map it is centred at galactic $(l, b) =$ (203ř, −56ř), with a decrement of $-70\,\mu$K at 5ř smoothing and a peak of $-168\,\mu$K at 2ř smoothing, and a half-maximum radius of 2.5ř; we use these values throughout.

Cyclonic vortex reading. Internal quantum turbulence in the bounce creates macro-vortices. In the **Jovian/cyclonic model** these act as thermal dams that choke local heat transport, deform under shear into elongated features aligned along a preferred rotation axis (the "Axis of Evil"), and — without solid-surface friction to dissipate angular momentum — scale up with expansion and freeze into the CMB at decoupling. The Cold Spot is read as such a frozen cyclone.

Failed prediction: the thermal ring. A cyclone diverting thermal energy around its core would produce a concentric hot ring of +20–50 μK at 5ř–7ř. The azimuthally averaged Planck 2018 SMICA profile ($N_{\rm side} = 128$) around the Cold Spot, with uncertainties from 500 random high-latitude positions (Fig. 5), shows a cold core extending to $\sim$9ř (-3.4σ at 3ř) and **no hot ring at** 5ř–7ř; a +37–52 μK excess at 13ř–15ř is $\sim 2.5\sigma$ before a trials factor of ~ 10 radial bins. The thermal-ring prediction failed (P1). The polarization part survives: a frozen vortex boundary should carry a radial E-mode ring of 1–3 μK at $\sim$5ř–15ř. Planck's Q_r profile is consistent with zero in every 2ř ring to 20ř with per-ring errors of 0.6–2.7 μK (Appendix A), so the ring is neither detected nor excluded; the test requires CMB-S4 sensitivity.

Preferred axis. Our simulations find ejection isotropic within shot noise (Sec. 3), so an axis can only be supplied by an external seed, the neighbouring bang of Sec. 7. The literature's cluster of tentative ($\lesssim 3\sigma$) alignments — a supernova acceleration dipole roughly along our CMB motion (Colin et al. 2019; Sah & Rameez 2024), the quadrupole–octupole alignment with the dipole direction,

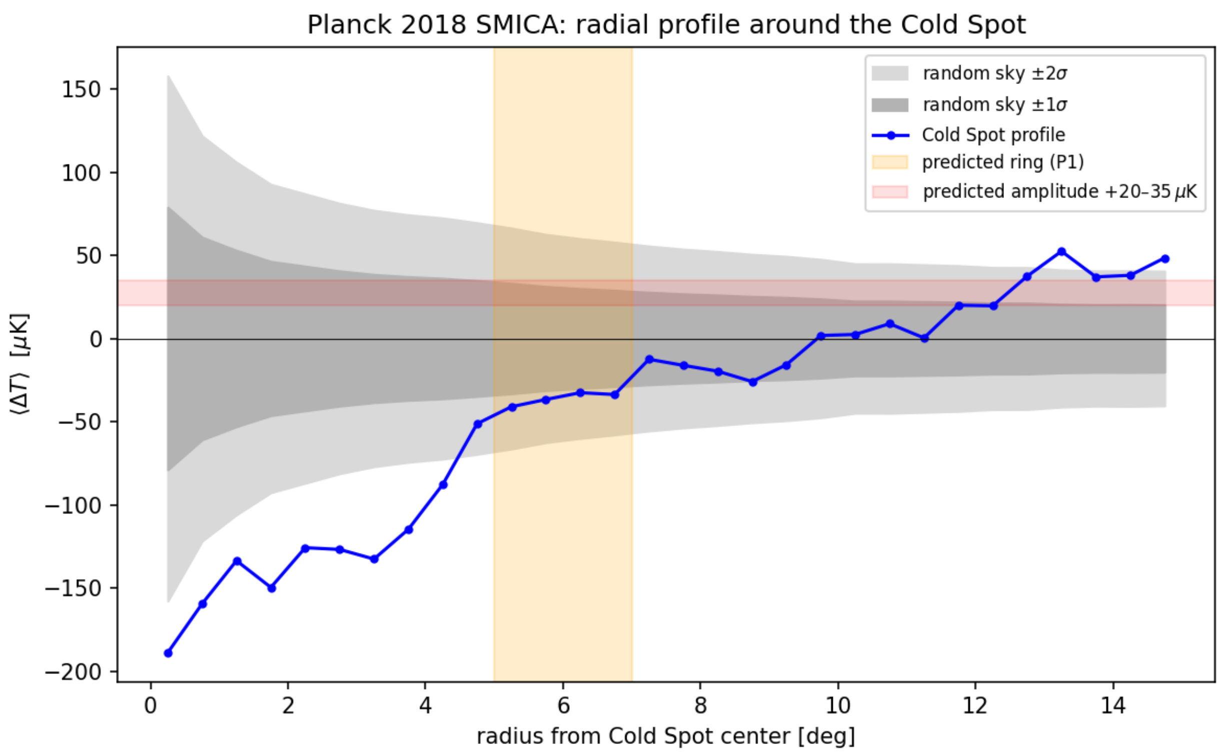

Figure 5: Fig. 5 — Planck 2018 SMICA radial temperature profile around the Cold Spot, vs. 500 random reference positions.

coincident bulk-flow and quasar-polarization axes (Hutsemékers et al.) — is motivation for H2/H3, not support.

Galaxy handedness. Claimed spin-handedness dipoles (Longo 2011; Shamir 2012–2022) lie 70ř from each other and 50ř–74ř from the axes above; we treat them as systematics-limited (Iye et al. 2021). The JWST/JADES handedness claim 8.6ř from the Cold Spot is a coincidence awaiting a control-field test (Appendix A).

Falsifier. Absence of the E-mode ring at CMB-S4 sensitivity eliminates H2. A localized curl detection (P8) at the Cold Spot would support the vortex reading over the neighbour reading of the same direction (Sec. 7); a calibrated curl null retires it (Table 2).

7. H3 and H4: A Neighbouring Big Pool Bang in the Cold Spot Direction, and a Population of Neighbours

Because vacuum crystallization is stochastic across an unbounded background, our bang need not be unique. A neighbouring universe nucleating beyond our domain boundary imprints our last-scattering surface, at comoving distance $d_{\rm LS} \approx 46$ Gly (4.35×10^{26} m); a neighbour closer than that would appear in large-scale-structure tracers, which the Cold Spot sightline does not show beyond the too-shallow Eridanus supervoid.

Causal admissibility. A source at $D = (1+\epsilon)\,d_{\rm LS}$ with $\epsilon > 0$ lies beyond our particle horizon. Nothing emitted by it since the bounce — no black holes crossing the boundary, no gravitational-wave memory, no tidal torque acting over the past Gyr — can have reached our volume. The only causally admissible imprint is one already present in the initial conditions: a potential Φ of the Grishchuk–Zel'dovich type laid down at the bounce, which the last-scattering photons sample as a Sachs–Wolfe spot with a $1/r$ tail over the whole sky at low multipoles. The neighbour reading therefore makes two kinds of prediction: the spot itself (amplitude, profile, non-rotation, polarization edge) and its low-ℓ tail (P9). Its present-day effect on our volume is a static tidal field fixed once the mass and distance are (Sec. 8).

The Sachs–Wolfe model. For a static potential imposed on the initial conditions, $\Delta T/T \approx \Phi/c^2$ with coefficient ≈ 1; the familiar 1/3 applies to adiabatic growing modes, not to an imposed external potential. At the spot centre $\Phi = GM/(\epsilon\, d_{\rm LS})$, and the angular profile is

$$\Delta T(\theta) \propto \Big[\epsilon^2 + 2(1+\epsilon)(1-\cos\theta)\Big]^{-1/2},$$

so the half-maximum angle fixes the distance, $\epsilon = \theta_{1/2}/\sqrt{3}$, and the amplitude then fixes the mass, $M = (\Delta T/T)\,\epsilon\, d_{\rm LS}\, c^2/G$. Because $\theta_{1/2}$ is beam- and smoothing-limited at 2ř smoothing, the inferred ϵ values are upper bounds. The mass function of a population of such spots is the nucleation size spectrum of the vacuum transition — the distribution that drove the cascade of Sec. 1 — so the internal (ejecta) and external (spot) samples of one process must agree, a cross-consistency requirement linking H1 and H4.

Southern one-sidedness. All eight most-extreme large-scale spots on the SMICA map ($|b| > 25$ř, 2ř smoothing; Appendix A) lie in the southern galactic hemisphere; in 200 Gaussian realizations through the same pipeline none placed its full top-8 in one hemisphere (mock mean 5.2 of 8). The significance to be quoted is that of the published hemispherical power asymmetry ($p \sim 1$–10%, estimator-dependent), since our mocks are simplified and the frame post hoc. In this framework a single neighbour to the galactic south modulates large-scale power on one side of the sky through its initial-condition potential, so the asymmetry, the Cold Spot, and the preferred axis of Sec. 6 become one prediction with one direction. An isotropic swarm of neighbours (H4) predicts no such one-sidedness.

H4. Since nucleation is stochastic, the neighbour should not be unique: a population of Big Pool Bangs at varying distances, each imprinting a distinct anomaly (P6). Planck 2018 isotropy analyses find the sky consistent with Gaussian ΛCDM apart from 2–3σ large-scale anomalies and the single Cold Spot, so additional neighbours must imprint below Planck's per-spot threshold. H4 stands or falls on population statistics (Sec. 9), not on any single spot.

Falsifiers. H3 falls to a foreground structure accounting for the full decrement, a curl detection at the Cold Spot (which transfers the direction to H2), or a P9 null at the pre-registered threshold; H4 to a null in sub-threshold spot statistics at LiteBIRD/CMB-S4 depth (Table 2).

8. H6: Where Is the Second Big Pool Bang?

H6 states the neighbour's location once; other sections refer here.

Direction. The Cold Spot, $(l, b) = (203ř, -56ř)$, uncertain by $\sim$2–3ř (the half-max radius), in the hemisphere of the eight-spot one-sidedness.

Distance. With $\theta_{1/2} = 2.5ř$ the profile of Sec. 7 gives $\epsilon = 0.025$; the eight extremes span $\epsilon = 0.025$–0.055 (Sec. 9). The source lies $\epsilon\, d_{\rm LS} \approx 1.1$ Gly comoving beyond the last-scattering surface (up to 2.5 Gly for the widest spots), at $D \approx 47$ Gly $\approx 4.5 \times 10^{26}$ m; since $\theta_{1/2}$ is smoothing-limited, ϵ is an upper bound.

Mass. With $\Delta T = -168\,\mu$K (the 2ř-smoothing peak), $T = 2.725$ K, $\epsilon = 0.025$ and $d_{\rm LS} = 4.35 \times 10^{26}$ m,

$$M_2 = \frac{\Delta T}{T}\,\frac{\epsilon\, d_{\rm LS}\, c^2}{G} = (6.2 \times 10^{-5})\,(1.09 \times 10^{25}\ \mathrm{m})\,(1.35 \times 10^{27}\ \mathrm{kg\ m^{-1}}) \approx 9 \times 10^{47}\ \mathrm{kg} \approx 10^{48}\ \mathrm{kg},$$

uncertain by ± 0.5 dex from the range of ϵ (0.025–0.055) and of the amplitude (-70 to $-168\,\mu$K depending on smoothing): $\sim 4 \times 10^{-5}$ of our own $M_1 = 2.2 \times 10^{52}$ kg, a dwarf neighbour. We use $M_2 \approx 10^{48}$ kg throughout.

Tidal pull on the local volume. The neighbour's present-day acceleration of our volume is $GM_2/D^2 = (6.67 \times 10^{-11})(9 \times 10^{47})/(4.5 \times 10^{26})^2 \approx 3 \times 10^{-16}$ m s^{-2}; integrated over a Hubble time (4.4×10^{17} s) this is ≈ 130 m s^{-1}, i.e. $\sim$0.1 km s^{-1} and $\lesssim 1$ km s^{-1} after the mass uncertainty — negligible against the 419$\pm$36 km s^{-1} Cosmicflows-4 bulk flow (Watkins et al. 2023) toward $(298ř, -8ř)$, 86ř from the Cold Spot. The neighbour reading therefore predicts *no* bulk-flow contribution above $\sim$1 km s^{-1} (P10); the observed flow must be explained by local attractors. The differential field, $GM_2/D^3 \approx 7 \times 10^{-43}$ s^{-2}, is an expansion anisotropy of $\sim 10^{-7}\,H_0^2$, undetectable in supernova anisotropy searches and below the low-ℓ Grishchuk–Zel'dovich limit — a consistency check, not a discriminant. Hot spots require a boundary imprint of the opposite sign (an underdense or approaching wall); the model does not specify which neighbours imprint hot.

Our position relative to it. ≈ 47 Gly comoving along the axis toward $(203ř, -56ř)$; the internal offset of H5 ($\lesssim$ few Mpc, Sec. 10) is negligible by comparison.

Quantity	Estimate	Method	Status
Direction	(203ř, −56ř) ± 3ř	Cold Spot centroid; southern one-sidedness	Fixed by data; attribution hypothetical
Distance beyond last scattering	$\epsilon = 0.025$, ≈ 1.1 Gly comoving (upper bound)	$\epsilon = \theta_{1/2}/\sqrt{3}$	Smoothing-limited
Mass M_2	$\approx 10^{48}$ kg (±0.5 dex)	Sachs–Wolfe, coefficient 1, $\Delta T = -168\,\mu$K	Single adopted value
Induced local velocity (P10)	$\lesssim 1$ km s^{-1}	$GM_2/D^2 \times t_H$	Bound; flow is local
Expansion shear	$\sim 10^{-7}\,H_0^2$	GM_2/D^3	Consistent, undetectable
Curl at Spot (P8)	Zero (neighbour) vs excess (vortex)	Lensing curl / B-mode stack	Uncalibrated null; calibrated test pending
Low-ℓ tail (P9)	Fixed $\ell = 2, 3$ template	Correlation vs Gaussian null (Sec. 9)	Uninformative at present

What H6 predicts. A *non-rotating* imprint (zero curl at calibrated sensitivity), a disc-shaped edge with the Aguirre–Johnson bubble-collision polarization profile, a low-ℓ tail of fixed shape (P9), no bulk-flow contribution (P10), and no proper motion. Falsifiers are in Table 2 (Sec. 11); P8 is the decisive near-term test, separating the two readings of the same direction.

9. H7: The Population

(a) Universe 1 is one of several. Vacuum crystallization nucleated more than once; our bang is one of several separated by domain boundaries, each with its own furnace and ejected halo.

(b) Each neighbour imprints one spot. Under the Sachs–Wolfe model of Sec. 7, the eight most extreme large-scale spots on the SMICA map are the candidate neighbour list, each with $\epsilon = \theta_{1/2}/\sqrt{3}$ (an upper bound, since $\theta_{1/2}$ is smoothing-limited at 2ř):

Rank	(l, b)	Sign	ΔT (μK)	$\theta_{1/2}$	ϵ (upper bound)
1	(157ř, −71ř)	hot	+170	2.5ř	0.025
2	(80ř, −33ř)	cold	−170	3.5ř	0.035
3	(203ř, −56ř) (Cold Spot)	cold	−168	2.5ř	0.025
4	(155ř, −29ř)	cold	−163	5.5ř	0.055
5	(305ř, −29ř)	hot	+153	2.5ř	0.025
6	(211ř, −35ř)	cold	−153	2.5ř	0.025
7	(170ř, −47ř)	hot	+162	4.5ř	0.045
8	(184ř, −54ř)	hot	+155	4.5ř	0.045

The eight extremes require $\epsilon = 0.025$–0.055, i.e. sources 1–2.5 Gly (comoving) beyond last scattering. Under Sachs–Wolfe an overdense wall gives a *cold* spot; a hot spot requires an underdensity or an approaching (Doppler) wall, a sign mechanism the model leaves unspecified (Sec. 8). Spot 8 at (184.2ř, −54.3ř) lies 11ř from the Cold Spot centroid (203.2ř, −56.3ř) and may be part of one system; spot 7 at (170.3ř, −46.6ř) lies 22ř away and is separate. The candidate count is ≤ 8, or 7 if spot 8 is merged.

(c) Population predictions. Neighbours cluster on the sky (currently 8/8 south). Any genuine neighbour is non-rotating and should show zero curl; any relic vortex of our own bang (H2) should show P8 curl excess — the population divides on this test. The inferred neighbour mass function must match the nucleation spectrum inferred from our own ejecta. The exploratory locus, curl, lensing-convergence and tracer statistics on the eight spots are reported once, in Appendix A; none is significant.

(d) The low-multipole tail (P9). A neighbour that imprints a compact spot cannot imprint *only* a spot: the $1/r$ potential of each source extends over the whole sky at $\sim 2\%$ of its peak (3–6 μK at 90ř, 2–4 μK at the antipode). Summing the eight tails gives a fixed $\ell = 1$–3 template with rms 4.1 μK at $\ell = 2$ and 4.5 μK at $\ell = 3$, i.e. $\sim 20\%$ of the observed quadrupole power ($C_2 \approx 210\,\mu\mathrm{K}^2$) and $\sim 8\%$ of the octopole — allowed, and a prediction: the template must correlate positively with the observed $\ell = 2, 3$ maps. *Pre-registered threshold.* With only 5 ($\ell = 2$) and 7 ($\ell = 3$) real modes per multipole, the correlation coefficient r between independent Gaussian patterns has $\sigma \approx 0.4$–0.5 under the null, so a single value of r is uninformative. The test is: compute r between the template and the masked Planck Commander $\ell \leq 3$ maps, and the same r for 1000 Gaussian ΛCDM realizations (synfast, same mask and pipeline); a detection requires r above the 95th percentile of the null at *both* multipoles, and a value at or below the null median at both, with the template amplitude fixed by the spots, falsifies the population reading. An unmasked SMICA comparison ($r = +0.15$ at $\ell = 2$, -0.27 at $\ell = 3$; template axes 78ř and 52ř from the observed) is uninformative at this power and foreground-contaminated.

(e) One-sided or surrounding. A *surrounding* population would populate both hemispheres; the 8/8 southern count indicates a one-sided, clustered population, and H7 takes that as its picture. The symmetric lattice of Appendix B is the limiting symmetric case, whose cubic-harmonic ($\ell = 4$) expansion-rate signature is a further falsifier. The pre-registered extension is the hemispheric balance of the extremes ranked 9–28: a one-sided population predicts a persistent southern excess; a surrounding population, or a Gaussian sky with the top-8 count a fluctuation, predicts isotropy. What falsifies H7 as a whole is given in Table 2 (Sec. 11).

10. H5: Where Are We Relative to Our Own Big Pool Bang?

In the homogeneous picture (Sec. 3) ejection happens everywhere and there is no geometric centre, so H5 asks: **does any residual large-scale radial ordering survive in our Hubble volume?** Two signatures would answer yes: a displaced-centre dipole in the peculiar-velocity field, or a radial-ordering signature in the bulk-flow profile (a flow that persists rather than converges with depth).

Displacement from the centre. The machinery is standard (Alnes & Amarzguioui 2006; King & Ellis 1973; Asvesta et al. 2022). Attributing the full CMB dipole (370 km s^{-1}) to off-centring gives $\Delta r \approx 5$ Mpc; the residual unexplained by local gravity gives $\Delta r \lesssim 1$–2 Mpc. A displaced centre is invisible in a linear flow but enters at $\mathcal{O}(Hc^2/r)$, ~ 90 km s^{-1} at 10 Mpc for a 5 Mpc offset. Fitting the 2{,}112 Cosmicflows-4 groups at 1.5–40 Mpc with $v(R) = hR + kR^2$ diverging from a free centre gives an offset $|\boldsymbol{\xi}| = 6.1$ Mpc toward (118ř, −24ř) at $\Delta\chi^2 = 6595$, with shells at 40–80 and 80–150 Mpc returning centres within 10ř–15ř of it. Two controls dismantle this. *Mock mask:* ten isotropic mocks on CF4's exact sky positions and distance errors manufacture false centres (0.4–2.5 Mpc, $\Delta\chi^2 \approx$ 9–61 per shell) that in the outer shells point into the *same longitude band* ($l \approx$ 93ř–152ř) as the data. *Structure subtraction:* removing the 2M++ reconstruction-predicted peculiar velocities (Carrick et al. 2015) drops the inner shell to $\Delta\chi^2 = 182$ ($|\boldsymbol{\xi}| = 1.7$ Mpc, within 8ř of the CMB dipole, plausibly a residual of the reconstruction's external-dipole calibration) and the outer shells to the mock floor. **No displaced expansion centre is detected once mapped structure is subtracted**; any offset is $\lesssim$ a few Mpc, $\sim 10^{-4}$ of the distance to last scattering.

The bulk flow. The bias-corrected Cosmicflows-4 bulk flow (Watkins et al. 2023) is 419 ± 36 km s^{-1} at $200\,h^{-1}$ Mpc toward (298ř ± 5ř, −8ř ± 4ř), rising with depth, in nominal $\sim 4.8\sigma$ tension with ΛCDM (Whitford et al. 2023 argue the errors are understated); Planck's kSZ analysis limits coherent flows to $\lesssim 254$ km s^{-1} at 95% on larger scales, so amplitudes $\gtrsim 600$ km s^{-1} are disfavored (P4). The flow axis and the CMB dipole (264ř, +48ř) lie 40ř–50ř apart, which a single offset vector does not predict; local-attractor infall explains the flow without any offset, and the neighbour of Sec. 8 contributes $\lesssim 1$ km s^{-1}. The southern one-sidedness (Sec. 7) is 86ř from the flow axis and 116ř from the CMB dipole, so it is not an offset signature either.

Quantity	Estimate	Method	Status
Offset from centre	$\lesssim$ 2–5 Mpc; no detection	CMB dipole cap; CF4 free-centre fit with mock-mask and 2M++ controls	Null
Offset direction	Undetermined (CMB dipole (264ř, +48ř) vs flow (298ř, −8ř))	Dipole/flow attribution	Misaligned by 40ř–50ř
Bulk flow (P4)	419 ± 36 km s^{-1} at $200\,h^{-1}$ Mpc; $\gtrsim 600$ disfavored	Cosmicflows-4, kSZ	Persistence with depth untested

Falsifier and retirement condition. A detected displaced-centre dipole in the peculiar-velocity field (a structure-subtracted, mock-calibrated $|\boldsymbol{\xi}|$ above the mock floor with a stable direction across shells), or a bulk flow that persists rather than converges at bias-corrected depths $\gtrsim 200\,h^{-1}$ Mpc, would support H5. The current null is $|\boldsymbol{\xi}| \lesssim 2$ Mpc (the 1.7 Mpc structure-subtracted residual, itself plausibly a calibration artefact). If the next Cosmicflows release, with roughly twice the group count and a $\sim\sqrt{2}$ smaller offset error, returns a structure-subtracted limit $|\boldsymbol{\xi}| < 1$ Mpc — the CMB-dipole residual bound — together with a converging bulk flow, H5 has no observable signature at distance-catalogue depth and is retired as untestable at present.

11. Falsifiable Predictions, Instrumentation, and Falsification Criteria

Hypothesis hierarchy. H1 (bounce and ejection; P3, P5) is the core and must hold; H2 (vortex; P1, P2, P7, P8), H3 (neighbour; P8, P9), H4 (population; P6), H5 (our position; P4), H6 (neighbour location; P8, P9, P10) and H7 (population model; P6, P9) are separable: any can fail without falsifying the core mechanism, and each can hold without the others.

Table 1. Predictions.

#	Prediction (hypothesis)	Observable and required threshold	Instrument	Status
P1	Cold Spot ring (H2). Thermal ring of $+20$–$50\,\mu$K at 5ř–7ř: **failed** in Planck 2018 (Sec. 6). Surviving prediction: radial E-mode ring of 1–3 μK at 5ř–15ř	Polarization sensitivity $\lesssim 1\,\mu\mathrm{K}\cdot$ arcmin at $\ell \sim 20$–100	CMB-S4	Thermal: failed. Polarization: open
P2	Cyclonic B-mode pattern aligned with the Axis of Evil (H2)	Degree-scale B-modes, r-equivalent $\sim 10^{-3}$	CMB-S4, LiteBIRD	Open
P3	First-order-transition stochastic gravitational-wave background from bubble collisions in the crystallization (H1): broken power law peaked at the transition scale	Stochastic-background spectral shape at nHz–mHz	Pulsar timing arrays, LISA	Open
P4	Bulk flow of ~ 400 km s^{-1} persisting (not converging) at bias-corrected depths $\gtrsim 200\,h^{-1}$ Mpc toward $l \approx 298$ř (H5); $\gtrsim 600$ km s^{-1} disfavored by the Planck kSZ limit	Peculiar-velocity survey to $z \sim 1$ with $\lesssim 300$ km s^{-1} per cluster	Rubin/LSST + Euclid + kSZ	Data lean toward persistence; errors disputed
P5	No particle dark-matter candidate (DM = Planck relics) (H1)	Continued nulls approaching the neutrino floor	LZ, XENONnT	Ongoing

#	Prediction (hypothesis)	Observable and required threshold	Instrument	Status
P6	A sub-threshold population of Cold-Spot-class spots (H4, H7); Planck already limits any above-threshold population	Sub-threshold spot statistics; per-candidate polarization at $\lesssim 1\,\mu\mathrm{K}\cdot\mathrm{arcmin}$; hemispheric balance of extremes 9–28	LiteBIRD, CMB-S4; Planck (archival)	Constrained; open below threshold
P7	Spin surcharge (H2): filaments and clusters carry primordial angular momentum atop tidal spin, excess largest at high z, randomly oriented (zero net; $\omega/H_0 \lesssim 10^{-9}$). Wang et al. (2021) filament spin is the $z \approx 0$ anchor	Filament-spin stacking at $z \sim 1$ vs $z \sim 0$ against ΛCDM torque mocks	DESI, Euclid, 4MOST	Open
P8	Localized curl excess (lensing curl and B-mode variance) at the extreme-spot positions if they are relic vortices (H2); zero at the Cold Spot if it is a neighbour imprint (H3, H6). A calibrated stacked null at the Cold Spot retires the vortex reading (Appendix A)	Planck PR4 curl reconstruction at full depth; degree-scale B-modes at $\lesssim 1\,\mu\mathrm{K}\cdot\mathrm{arcmin}$	Planck PR4 (archival), CMB-S4, LiteBIRD	Uncalibrated null; calibrated test pending

#	Prediction (hypothesis)	Observable and required threshold	Instrument	Status
P9	Low-multipole tail (H3, H6, H7): the fixed $\ell = 1$–3 template summed from the eight point-mass profiles (rms $4\,\mu$K at $\ell = 2$, $4.5\,\mu$K at $\ell = 3$) must correlate positively with the observed $\ell = 2, 3$ maps; detection requires r above the 95th percentile of 1000 Gaussian realizations at both multipoles (Sec. 9d)	Masked Commander low-ℓ maps; r against the simulated null	Planck 2018/PR4 (archival)	Unmasked $r = +0.15$ ($\ell = 2$), -0.27 ($\ell = 3$): uninformative
P10	Bulk-flow bound (H6): the neighbour contributes $\lesssim 1$ km s^{-1} to the local bulk flow (Sec. 8); the observed flow must be accounted for by local attractors	Bulk-flow residual after 2M++/CF4 reconstruction $\ll 400$ km s^{-1} unexplained	CF4 (archival); DESI peculiar velocities	Bound; consistent

Table 2. Falsification criteria. This is the single reference for what each hypothesis needs and what removes it; Secs. 9, 12 and the Epilogue refer here.

Hypothesis	What supports it	What falsifies it	Current status
H1 bounce and ejection	$88\% \pm 4\%$ ejected fraction at $\sigma_{\rm rel} \approx 1$; particle-DM nulls (P5); a transition-shaped GW background (P3)	A particle dark-matter detection; an ejecta velocity spectrum that cannot yield a cold matter component by matter–radiation equality; ρ_c inconsistent with the LQC bound	Open; relativistic formulation and ejecta temperature unresolved

Hypothesis	What supports it	What falsifies it	Current status
H2 Cold Spot vortex	E-mode ring (P1), cyclonic B-modes (P2), curl excess at the Spot (P8), spin surcharge (P7)	Absence of the E-mode ring at CMB-S4 sensitivity; a calibrated stacked curl null at the Cold Spot	Thermal ring failed; polarization open; curl uncalibrated
H3 neighbour in the Cold Spot direction	Zero curl at the Spot (P8); positive low-ℓ tail correlation (P9); bubble-collision polarization edge	A foreground structure accounting for the full decrement; a curl detection at the Spot; P9 at or below the null median at both multipoles	Open; P9 uninformative
H4 population of neighbours	Sub-threshold spot population (P6); mass function matching the ejecta nucleation spectrum	Null sub-threshold spot statistics at LiteBIRD/CMB-S4 depth	Constrained above threshold; open below
H5 residual radial ordering	Displaced-centre dipole above the mock floor; bulk flow persisting with depth (P4)	Structure-subtracted $\|\boldsymbol{\xi}\| < 1$ Mpc in the next Cosmicflows release with a converging flow (retired as untestable)	Null; $\|\boldsymbol{\xi}\| \lesssim 2$ Mpc
H6 location of the second bang	Non-rotating imprint (P8); low-ℓ tail (P9); no bulk-flow contribution (P10); Aguirre–Johnson polarization edge	Curl detection at the Spot; absence of the bubble-collision polarization profile at CMB-S4 sensitivity; one-sidedness shown to be an estimator or foreground artefact in end-to-end simulations	Open

Hypothesis	What supports it	What falsifies it	Current status
H7 one-sided population	Southern excess persisting in extremes 9–28 (P6); positive P9; zero curl at all eight spots	All of: P9 failing at the pre-registered threshold; zero curl at calibrated sensitivity at all eight spots *and* no bubble-collision polarization profile at any at CMB-S4 depth; no CMB-independent mass counterpart in a pre-registered galaxy-density re-run. For the lattice limit: a detected $\ell = 4$ cubic expansion-rate pattern (Appendix B)	Open; 8/8 south at $p \sim 1$–10%

Under the joint nulls of H3, H4, H6 and H7 the spot population is Gaussian and the Cold Spot a single outlier of unknown origin, while H1, H2 and H5 stand. The polarization ring and calibrated curl test await Simons Observatory / CMB-S4 (a 2030s timeline); P9 and P10 are archival.

12. Limitations and Open Problems

1. **Hard-sphere ansatz.** Treating Planck relics as incompressible spheres is a modeling assumption motivated by the Compton–Schwarzschild coincidence, not a quantum-gravity result.
2. **Bounce mechanism and horizon structure.** The rebound requires the LQC-form modified Friedmann equation; $\rho_c = \eta_{\rm rcp}\rho_P$ is a conjecture. The bounce is meaningful only for a whole or super-horizon patch, and a trapped-surface timing analysis, together with the multi-bubble assembly timescale ($\gtrsim 10^{20}\, t_P$), remains to be done.
3. **Spectral index.** ν_t and the relation $n_s - 1 = -\nu_t/3$ are fitted inputs; the compatibility of a Kolmogorov cascade with near scale invariance is unproven.
4. **Dark-matter fraction and temperature.** The 88% ± 4% ejection requires $\sigma_{\rm rel} \approx 1$, which the cascade model reaches only marginally, and the end-to-end chain is untested. The ejecta are relativistic, hence hot rather than cold dark matter unless redshifting or cooling reduces their free-streaming length by matter–radiation equality. Planck-relic dark matter must also face femtolensing, capture, and abundance constraints.
5. **Expansion history.** Neither the Newtonian toy nor the LTB lightcone produces apparent acceleration, the toy's $w(z)$ drift runs opposite to the DES+DESI direction, and the kSZ gate caps any ejecta contribution at $\sim$1% of dark energy. The adopted history (Sec. 5.2) is the pool model plus a constant tension on an assumed flat ΛCDM background; the tension's value is not predicted. The model is also not yet reconciled with BBN abundances or the CMB acoustic peaks.

6. **Causal and relativistic structure.** Homogeneity of the $\sim 10^{20}\,\ell_P$ patch must be assumed (the horizon problem stands). Ejection proceeds where Newtonian escape velocity vastly exceeds c; the N-body mechanism lacks a relativistic formulation, prerequisite for H1 to be a physical theory rather than an analogy. The neighbour of H3/H6 lies beyond the particle horizon, so its imprint must be an initial-condition potential (Sec. 7); how such a potential is laid down across a domain boundary is not modelled.
7. **Relic-to-plasma conversion.** The furnace must convert relics to Standard-Model plasma with the observed baryon asymmetry while ejected relics stay stable; no channel is specified.
8. **Sign of hot spots.** Sachs–Wolfe supplies cold spots from overdense walls; the hot extremes require an unspecified underdensity or Doppler mechanism (Sec. 8).

Each item maps onto the observational program of Sec. 11 (Tables 1 and 2); the theory is offered as a falsifiable mechanical alternative, not a completed replacement for ΛCDM.

Data and code availability. Code, figures, build instructions and download links for all datasets used are at https://github.com/AIMFIRST-VN/big001.

Epilogue: Three Questions, Three Answers

We set out with three questions and a pool table. Here is where the balls came to rest.

How does it work? Picture the moment before the first moment: a false vacuum, tense as a drawn bow. Somewhere a bubble of true vacuum nucleates, its wall sweeping outward, catalysing more bubbles, until the whole patch is a jammed crush of 10^{60} Planck-mass spheres pressed to the random-close-packing limit. Nothing can fall further. A loop-quantum bounce fires, the pile rebounds, and the rebound is chaotic: $88\% \pm 4\%$ of the spheres are hurled outward, the fastest furthest, sorted by speed like shot from a gun. That flying debris is our dark matter, and its velocity ordering is our Hubble flow. But the debris does not push the universe faster; nothing outside a volume ever can, and nothing inside it pushes either. The acceleration we measure comes from a constant tension of space, $w = -1$, which we adopt on an assumed ΛCDM background and cannot yet derive. So the model works as far as the matter goes, hands the expansion to a tension it does not explain, and carries three confessed wounds: the ejecta are born relativistic and must somehow cool, the thermal ring around the Cold Spot is not there, and the bounce is only meaningful if the jammed patch was the whole universe rather than a ball inside its own horizon.

Where are we? In the homogeneous picture the rebound happened everywhere, so there is no centre to be far from; the question that survives is whether any large-scale radial ordering is left in our patch. We looked for our own displacement from an expansion centre in the galaxy velocity field, and after the mock-mask and 2M++ controls the answer is: no measurable offset, two megaparsecs at most, direction undetermined. What we do have is a bulk flow of about 400 km s^{-1} that the kinetic Sunyaev–Zel'dovich data cap near 600, and a sky whose extreme spots all lie in the southern hemisphere. The reading we adopt is that we are not visibly off-centre, that the lopsidedness of the sky is more plausibly the work of a neighbour than of our own position, and that if the next velocity catalogue pushes the offset below a megaparsec the question is retired as untestable (Table 2). Where we are remains the least answered of the three questions.

Where is the second Big Bang? Follow the coldest patch of the oldest light: galactic longitude 203ř, latitude −56ř, the Cold Spot. If it is the fingerprint of a neighbouring Big Pool Bang written into the initial conditions of our last-scattering surface, then that neighbour sits just beyond it, about one billion light-years past the edge of everything we can see, with a mass of order 10^{48} kg — a dwarf beside our own 2×10^{52}. It lies beyond our horizon, so nothing it has done since can reach us: no black holes, no gravitational-wave echoes, and a tidal pull on our neighbourhood of a tenth of a kilometre per second, too small to move anything we can measure. Seven other extreme spots line up on the same side of the sky, a candidate pool of neighbours clustered rather than surrounding us. The neighbour reading makes promises it can be held to. It must leave no curl-mode fingerprint at the spot, where a vortex would (P8). Its $1/r$ potential must leak a few microkelvin across the whole sky in a pattern fixed by the spots (P9), to be judged against a thousand Gaussian skies rather than by eye. And it must contribute nothing to the bulk flow (P10), which it does not. The neighbour hypothesis will not survive as a story; it will survive, or fall, as the set of numbers in Table 2.

That is the adventure in one paragraph: a jammed crystal that bounced, a velocity-sorted debris field we live inside, and a cold fingerprint on the far wall that may belong to someone else. The next moves are already on the table.

Appendix A. Exploratory Null Results and a Pre-registered Test

This appendix records the exploratory battery run on the eight most extreme large-scale spots of the Planck SMICA map ($N_{\rm side} = 128$, $|b| > 25$ř, 2ř smoothing; per-spot amplitude ΔT and half-maximum radius θ; positions in Sec. 9). Nulls are 60–300 Gaussian realizations drawn from the map's own pseudo-C_ℓ and passed through the identical pipeline unless stated otherwise.

Locus test. If neighbours shared a common distance shell and a common formation density, their spots would follow a locus $|\Delta T| \propto \theta^3$, whereas Gaussian peaks obey BBKS statistics with amplitude and size nearly independent. A log–log fit of $|\Delta T|$ against θ gives slope -0.00, $r = -0.01$, 27th percentile of the Gaussian null ($+0.07 \pm 0.10$). No amplitude–size correlation is detected, but the test is powerless as implemented (radii quantized to 2.5ř–5.5ř, amplitude spread $\sim$10%). It is reported as unresolved pending injection-recovery validation, not as an exclusion. Neighbours at a range of distances would populate a diffuse cloud rather than a locus, and one genuine neighbour among seven Gaussian impostors would not move the fit.

Internal dipoles and spin budget. Monopole-plus-dipole fits within 6ř in the local tangent frame: the eight extremes split 4/4 hot/cold with rank-matched amplitudes summing to $-11\,\mu$K against a mean $|\Delta T| \approx 162\,\mu$K (7% of Gaussian skies as balanced); opposite-sign spots lie 4.0ř closer than same-sign (null +5.2ř ± 16.3ř, $p = 0.30$); nearest-neighbour spacing is no more regular than Gaussian. Converting dipoles to spin-axis proxies ($\mathbf{L} \propto \hat{n} \times \mathbf{d}$), the Cold Spot's internal dipole ($16.4\,\mu$K/deg) is the largest by a factor 2.3 and the others weakly co-align. These fits are dominated by $\ell \sim 2$–20 gradients that Gaussian peaks generically carry and that are correlated across co-hemispheric spots, so they are upper bounds on rotation, not spin measurements; a gradient-subtracted re-fit is required before any spin fraction can be quoted.

Spiral winding. The azimuthal phase of the $m = 1$ mode in 1.5ř rings from 1.5ř to 15ř around the Cold Spot is locked ring-to-ring with coherence 0.988 — above all 300 random positions (null 0.72 ± 0.18) — and drifts by $\sim$ 3ř of phase per degree of radius ($\sim$ 40ř total twist). A static background gradient gives high coherence but zero winding; only the winding is spin-specific. The $m = 2, 3$ modes are unremarkable, and this is a post-hoc test.

Curl search. The public PR4 lensing release contains only gradient modes, so we implemented an approximate temperature-only quadratic estimator ($N_{\rm side} = 256$, $\ell \leq 512$, spin-1 curl projection, mean field from 40 Gaussian simulations; unnormalized, valid only as a relative comparison). Stacking curl-potential variance in 6ř discs: the eight spots sit at the 82nd percentile of 60 simulated skies and the Cold Spot at the **38th** — indistinguishable from a random position. This does not trigger the retirement clause of P8, which is bound to a sensitivity reaching the temperature-dipole-implied rotation that this reduced-resolution estimator does not attain (Planck's full-depth reconstruction uses $\ell \leq 2048$ with polarization).

Polarization. On the SMICA Q/U maps ($N_{\mathrm{side}} = 256$, $\ell \leq 512$, E/B decomposition, 1ř smoothing): stacked B-mode variance in 6ř discs at the eight spots is at the 66th percentile of random positions, E-mode variance at the 35th, and the Cold Spot's radial-E profile is consistent with zero in every 2ř ring to 20ř with per-ring errors of 0.6–2.7 μK — the predicted 1–3 μK ring (P1) is neither detected nor excluded.

Lensing convergence. The sign-weighted PR4 κ stack on the eight positions is $+0.006$ against a random-position null of 0.000 ± 0.013 (4/8 sign matches): no integrated mass scars. The Cold Spot's inner-2ř $\kappa = -0.070$ vs null $+0.013 \pm 0.042$ (-2σ) has the sign of its known shallow foreground supervoid; the 8ř–20ř annulus never exceeds $+0.6\sigma$, so there is no mass "belt". A stack on $\sim$27,000 ordinary small spots (1ř smoothing, $|T| > 1.5\sigma$) shows cold spots on κ deficits (-7.2σ) and hot on excesses ($+2.2\sigma$), but κ is reconstructed from the same temperature data and quadratic estimators leak temperature extrema into the mass map; whether this is leakage or genuine ISW×lensing is undiscriminated, and a polarization-only κ stack is the required discriminating check.

CMB-independent tracers. Against the Quaia Gaia–unWISE quasar catalogue (756k quasars, $z \sim 1$–3, selection-function corrected) the small-spot correlation vanishes (cold -1.1σ, hot -0.3σ) and the eight extremes are within $\sim 1\sigma$ of random (Cold Spot -0.018, 0.6σ); this is not a sensitive null, since Quaia's kernel barely overlaps the ISW kernel. Against WISE×SuperCOSMOS (17.7M galaxies, $z = 0.1$–0.4, high-passed at 10ř): small spots remain null (cold -0.7σ, hot $+0.2\sigma$), while the extremes show seven of eight sign matches and a sign-weighted stack of $\approx +2.7\sigma$ against a correlated-noise null from 300 random 5ř discs. Qualifications: the result rests on three sightlines ($|z| \approx 2.0$–2.3); the nominal binomial 3.5% ignores the three-tracer, multi-statistic search (global $p \sim 10$–30%); extremeness-conditioning inflates the recovered correlation (Eddington-type bias); the 10ř high-pass self-subtracts genuine $\gtrsim$ 10ř supervoids; and extinction/stellar residuals at $|b| = 25$–40ř are of order the per-spot signals. Verdict: a 1–2σ-class hint that the extremes are partly sourced by $z \sim 0.1$–0.4 superstructures, to be confirmed or retired by a pre-registered re-run at two baseline scales with $|b| > 30$ř and extinction regression. The 2M++ cross-match of the eight sightlines ($< 180\,h^{-1}$ Mpc) matches spot signs at coin-flip rates (5 of 8).

JWST handedness. The JADES field at (224ř, -54ř), 8.6ř from the Cold Spot, shows a claimed 2 : 1 handedness asymmetry in 263 high-redshift galaxies (Shamir 2025). The sample is small, the method contested, and a Doppler boost from the Milky Way's rotation is a mundane candidate. The discriminating measurement is a control field — the identical pipeline on COSMOS or GOODS-N ($>$ 100ř away): a systematic reproduces the asymmetry everywhere; a Cold Spot spin column produces it only on the column. The local anchor is already measured: the same-handedness angular correlation over 77{,}840 SDSS galaxies (11.2M pairs, 0.05ř–4ř) is zero in every bin, with a 1.9σ global excess.

Coherence-gradient hypothesis (untested). In the N-body toy, only the earliest, fastest ejecta escape in a single pass and preserve geometric correlations; everything slower undergoes violent relaxation, which conserves spin magnitude but randomizes spin geometry. If any residual radial ordering survived in our Hubble volume (H5), one would expect a *coherence gradient*: fossil geometry intact in the earliest-ejected material, degrading toward the well-mixed later ejecta. Intermediate tracers (quasar-polarization alignments at $z \sim 1$–2; galaxy handedness axes) lie 24ř–73ř from the

Cold Spot direction, consistent with random axes, and the local same-handedness correlation above is zero to a few parts in 10^3, as mixing requires. This is recorded as a hypothesis; it has no validated observable and is not part of the main-text claims.

Statistical accounting. Across the battery (locus slope, hemisphere count, hot/cold balance, pairing distance, lattice regularity, dipole magnitudes, cancellation ratio, winding coherence, κ stack, curl percentile, tracer stacks) the individual results are $p \sim 0.06$–0.7; under any global look-elsewhere correction the sub-2σ items are jointly consistent with noise. Further method limits: the Gaussian nulls use the unmasked pseudo-C_ℓ without mask deconvolution, anisotropic noise, or foreground residuals (end-to-end NPIPE simulations are the required upgrade for any sub-percent claim), and the spots were selected on the same map on which all statistics were evaluated, so the hypotheses were partly formulated post hoc. What survives is qualitative: the extreme-spot sky is one-sided at literature-level significance, the Cold Spot is the outlier in every channel examined, and nothing requires a population of additional objects.

Pre-registered commitment (P8). The load-bearing, falsifiable claim of the vortex reading is the curl test. We commit in advance that a clean stacked curl-mode null at the Cold Spot position, at a sensitivity reaching the temperature-dipole-implied rotation, *retires* the vortex interpretation of that direction in favor of the non-rotating neighbour reading (H3/H6) — not merely disfavors it. If curl variance is instead detected localized there, the vortex reading stands and the Cold Spot becomes a relic vortex of our own bang rather than a neighbour's imprint.

Appendix B. A Fixed Grid of Surrounding Bangs (Consistency Check)

The population model in its most symmetric form — the limiting case of the one-sided population H7 adopts (Sec. 9e) — places Big Pool Bangs on a fixed lattice with our own at one site. As a consistency check, we ask whether the neighbours' pull on our ejecta could contribute to the observed late-time acceleration. Point masses of equal mass were placed on a cubic grid of unit spacing (26 neighbours for $3 \times 3 \times 3$, 124 for $5 \times 5 \times 5$); test galaxies were placed at distance r from us in random directions and the vector sum of the neighbours' inverse-square pulls was evaluated. Units: the pull of one neighbour on us.

r (grid units)	Sky-averaged outward pull	Toward a neighbour (axis)	Toward a gap (diagonal)
0.1	0.0000	+0.012	−0.008
0.2	0.000	+0.10	−0.06
0.3	0.000	+0.34	−0.20
0.4	0.00	+0.85	−0.44
0.45	0.00	+1.26	−0.60

The results are identical for both grid sizes. Galaxies lying toward a neighbour are pulled outward with a pull that grows steeply with r; galaxies lying toward the gaps between neighbours are pulled inward; the sky average is zero at every radius. This is Gauss's law, not a property of the arrangement: the sky-averaged radial acceleration at radius r is $GM_{\text{enc}}(r)/r^2$, and external sources enclose nothing. Only mass, or negative pressure, *inside* the observed volume can change the isotropic expansion rate. The lattice therefore predicts (i) zero contribution to the monopole acceleration, and (ii) a direction-dependent expansion rate with the cubic-harmonic ($\ell = 4$) pattern of the lattice, fast toward the six nearest neighbours and slow toward the eight diagonal gaps. Observation (ii) is a further falsifier for the lattice limit of H7 (Table 2): supernova and BAO expansion-rate anisotropy is limited to the few-percent level with no cubic pattern, while the amplitude implied by the neighbour mass and distance of Sec. 8 is $\sim 10^{-7}$ at the $r \approx 0.3$ reached by the supernova sample. The one form in which surrounding mass does raise the local expansion rate isotropically is a local underdensity, where the effect comes from the mass *missing inside*. A $\delta \approx -0.3$ underdensity within $\sim$300 Mpc, as claimed by Keenan, Barger & Cowie (2013), would imply an outflow of $\sim$1000 km s^{-1} at its edge, in tension with the kSZ gate of Sec. 5; the literature is divided (Keenan–Barger–Cowie 2013 for; Wu & Huterer 2017 and Kenworthy, Scolnic & Riess 2019 against), and this paper takes no position on whether the void exists. What it does conclude is that such a void cannot be the acceleration: a Gpc-scale void deep enough to mimic Λ is excluded by the kSZ gate. This appendix is a consistency check on the adopted expansion history of Sec. 5.2, not its basis.

Appendix C. Kinematic Mimicry of Acceleration: Tests

This appendix records the tests summarized in Sec. 5 of whether the ejecta distribution could mimic acceleration for an interior observer.

Relativistic (LTB) recalculation. Dust shells in the Lemaître–Tolman–Bondi metric obey the same radial equation as Newtonian shells; what the toy of Sec. 5.1 gets wrong is the *observables.* Integrating radial null geodesics through the evolving LTB metric (300 shells seeded from the ejecta speed distribution, fallback mass virialized into the interior, $d_L = (1+z)^2 R$) gives $q_0^{\rm eff} = +0.16$ ($w_{\rm eff}(0) \approx -0.23$): mild deceleration, no apparent acceleration, with a distance-modulus shape $\sim$0.35 mag from both ΛCDM and coasting out to $z \sim 5$. Scope restrictions: the lightcone threads only the free-streaming 9%, the interior treatment and observer epoch are modelling choices, and shell crossing was not monitored.

Profile inversion. Since the ejection profile is an input, we searched power-law, bimodal fast-tail, and stochastically lumpy profiles for any that accelerates. Smooth families robustly decelerate ($q_0^{\rm eff} \sim +1$ to $+4$); apparently negative values in lumpy configurations are seed-ensemble noise (± 1.4, exceeding the target $|q_0| \approx 0.55$, so that family is unconstrained rather than excluded). The known way an LTB model mimics acceleration — the giant void, observer inside a Gpc-scale underdensity — is the *opposite* density ordering from outward ejection.

Blast wave and the kSZ gate. A rebound acting as a blast wave would sweep ejecta into a shell and leave a rarefied cavity, the giant-void geometry; swept-shell profiles do give smoother lightcones (0.03–0.08 mag residuals vs $\sim$1 mag). This is not a rescue: the comparison metric cannot separate ΛCDM from coasting (~ 0.04 mag rms apart), a 0.08 mag residual corresponds to $\Delta\chi^2 \sim 400$ against Pantheon+, a cavity deep enough for $q_0 \approx -0.55$ is the giant-void model excluded by kinetic Sunyaev–Zel'dovich and Compton-y constraints (Zhang & Stebbins 2011), and mass swept into a distant shell is unavailable as local dark matter. Imposing the constraints first: the kSZ limit on coherent cluster flows ($v \lesssim 300$ km s^{-1} at Gpc distances) caps the mean interior contrast of any cavity at $|\bar{\delta}| \lesssim 0.05$ (500 Mpc) to $\lesssim 0.01$ (3 Gpc), i.e. $|\Delta q_0| \lesssim 0.01$ against the $\Delta q_0 \approx -1.05$ required. No ejection profile of any morphology supplies more than $\sim$1% of the observed dark energy; this is the result quoted in Sec. 5.

A framework-native candidate, with its cost. If crystallization is incomplete today, a residual unconverted vacuum fraction would be smooth, non-clumping, and immune to the kSZ gate — qualitatively dark energy. Quantitatively this re-imports the cosmological-constant problem verbatim ($\rho_P \sim 10^{96}$ vs 6×10^{-27} kg m^{-3}: tuning to one part in 10^{123}), and ongoing conversion would imply present-epoch nucleation, domain walls (Zel'dovich–Kobzarev–Okun bound), and an unconverted component at BBN and recombination — none computed here. The boiling picture does anchor H3–H4 in Coleman–De Luccia bubble nucleation, whose collisions imprint disc-shaped CMB features with specific polarization profiles (Aguirre & Johnson) and, generically, source the stochastic gravitational-wave background of P3.